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sap reaches the protoxylem a pressure is developed known as **root pressure**. This theory involves the symplastic movement of water.

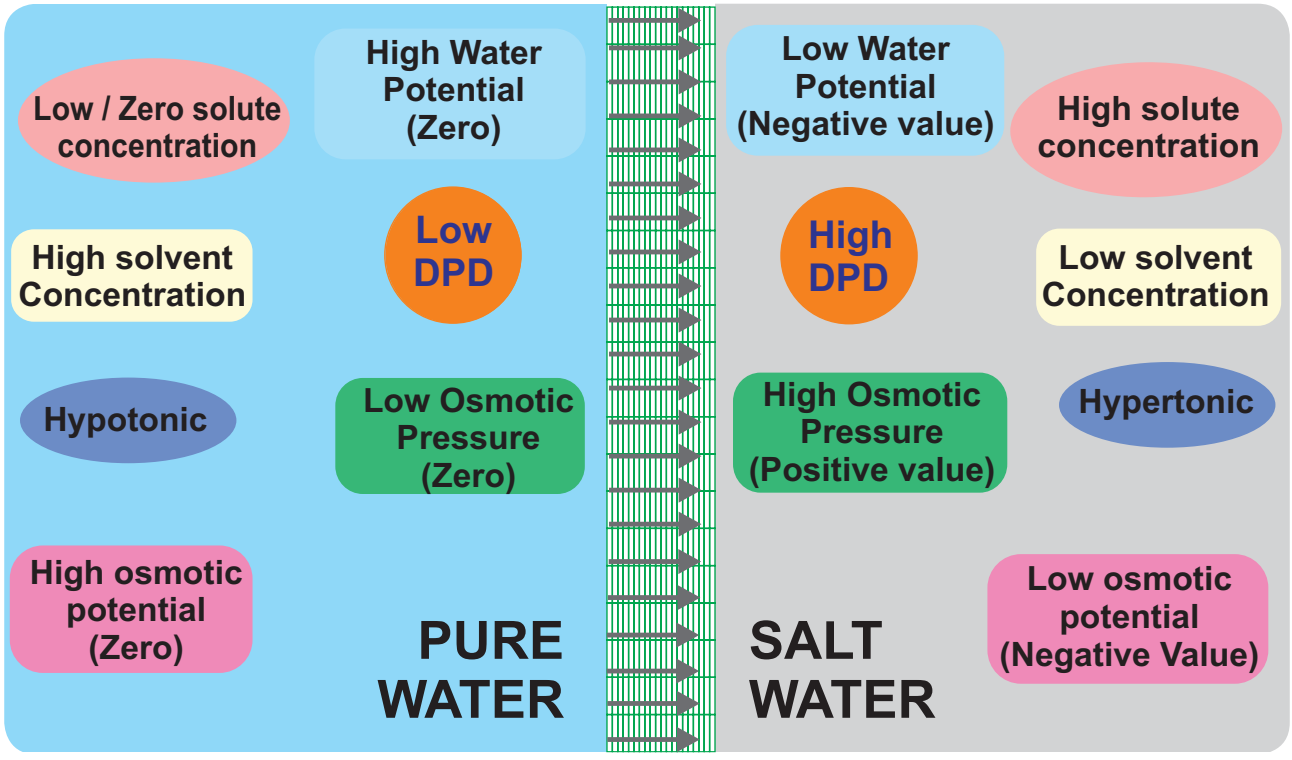
Objections to osmotic theory: 1.The cell sap concentration in xylem is not always high. 2. Root pressure is not universal in all plants especially in trees.

ii. Non-Osmotic active absorption

Bennet-Clark (1936), **Thimann** (1951) and **Kramer** (1959) observed absorption of water even if the concentration of cell sap in the root hair is lower than that of the soil water. Such a movement requires an expenditure of energy released by respiration (ATP). Thus, there is a link between water absorption and respiration. It is evident from the fact that when respiratory inhibitors like KCN, Chloroform are applied there is a decrease in the rate of respiration and also the rate of absorption of water.

2. Passive Absorption

In passive absorption, roots do not play any role in the absorption of water and is regulated by transpiration only. Due to transpiration, water is lost from leaf cells along with a drop in turgor pressure. It increases DPD in leaf cells and leads to withdrawal of water from adjacent xylem cells. In xylem, a tension is developed and is transmitted downward up to root resulting in the absorption of water from the soil.

In passive absorption (Table 11.3), the path of water may be symplastic or apoplastic. It accounts for about 98% of the total water uptake by plants.

Concept Map - Movement of water in an osmotic system based on various parameters